Euler Universal Phase Constraint: Formalism, Stabilization Mechanisms and Numerical Realizations (3-level system & Sagittarius A*)

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Abstract

We present a unified formalism for a global phase consistency identity — the *Euler Universal Phase Constraint* — which sums known quantum and topological phases into a single operator condition. We introduce a hierarchy of stabilization mechanisms (Riemann Zeta modulation, discrete Collatz-like iteration, Twin Primes grid, Ramanujan sum discretization, fractal stability filters, Operator of Intention and White Threads) and demonstrate how they can yield topological stability for model microstate ensembles. Numerical derivations for a 3-level quantum system and a stylized microstate model for Sagittarius A* are provided. We discuss emergent vs. imposed ingredients and give an operator-level account for Black Hole / White Hole duality in the phase framework.

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1 Introduction

This manuscript develops and documents the Euler Universal Phase Constraint (henceforth *Euler constraint*) and a practical stabilization pipeline that ensures global phase coherence for systems whose physical description includes geometric, dynamic and topological phases. The construction is intentionally modular: start from well-known phase contributions, then layer stabilizers that select discrete, topologically robust summations.

Motivation:

- unify Berry, Dirac/Aharonov–Bohm, Pancharatnam, dynamical, Zak and Aharonov–Casher phases into a single operator constraint;
- provide numerically implementable and physically interpretable stabilization mechanisms (Zeta, Collatz-like iteration, Twin Primes, Ramanujan sums, fractal stability and White Threads);
- apply the formalism to a toy 3-level quantum system and to a stylized microstate ensemble representative of a supermassive black hole (Sagittarius A*), to examine implications for information flow (White Threads ∼ Hawking radiation analogue).

2 Core formalism: all phases and the Euler operator

We work in a Hilbert space $\mathcal{H}$ with states $|\Psi\rangle$. For each (micro)state n we list the phase contributions.

2.1 Phase definitions (per state)

For a single eigenstate (or effective microstate) $|n\rangle$:

Dynamical phase

$$\phi_{\text{dyn}}^{(n)} \;=\; -\frac{1}{\hbar}\int_0^T E_n(t)\,dt. \tag{1}$$

Berry (geometric) phase For an adiabatic closed path C in parameter space:

$$\gamma_B^{(n)}(C) \;=\; i\oint_C \langle n(\mathbf{R})|\nabla_{\mathbf{R}} n(\mathbf{R})|n(\mathbf{R})|\nabla_{\mathbf{R}} n(\mathbf{R})\rangle \cdot d\mathbf{R}. \tag{2}$$

Dirac / Aharonov–Bohm phase For charge q and vector potential $\mathbf{A}$:

$$\phi_D^{(n)} \;=\; \frac{q}{\hbar}\oint_C \mathbf{A}\cdot d\mathbf{l}. \tag{3}$$

Pancharatnam phase Between two (not necessarily orthogonal) states $|\psi_1\rangle, |\psi_2\rangle$:

$$\gamma_P \;=\; \arg\langle\psi_1|\psi_2|\psi_1|\psi_2\rangle\,. \tag{4}$$

Zak phase For 1D Bloch states $|u_k\rangle$:

$$\phi_{\text{Zak}} \;=\; i\int_{\text{BZ}} \langle u_k|\partial_k u_k|u_k|\partial_k u_k\rangle\; dk. \tag{5}$$

Aharonov–Casher phase Dipole $\boldsymbol{\mu}$ moving in electric field $\mathbf{E}$ along C:

$$\phi_{\text{AC}} \;=\; \frac{1}{\hbar c^2}\oint_C (\boldsymbol{\mu}\times\mathbf{E})\cdot d\mathbf{l}. \tag{6}$$

2.2 Global Euler operator

Define the single-state Euler phase operator (per microstate n):

$$\hat{\Phi}_{\text{Euler}}^{(n)} \equiv \hat{\phi}_{\text{dyn}}^{(n)} + \hat{\gamma}_B^{(n)} + \hat{\phi}_D^{(n)} + \hat{\gamma}_P^{(n)} + \hat{\phi}_{\text{Zak}}^{(n)} + \hat{\phi}_{\text{AC}}^{(n)}. \tag{7}$$

For the whole system (all microstates):

$$\hat{\Phi}_{\text{Euler}}^{(\text{total})} \equiv \sum_n \hat{\Phi}_{\text{Euler}}^{(n)}. \tag{8}$$

The **Euler Universal Phase Constraint** is imposed as:

$$\hat{\Phi}_{\text{Euler}}^{(\text{total})}\,|\Psi_{\text{total}}\rangle \;=\; 2\pi N\,|\Psi_{\text{total}}\rangle, \qquad N\in\mathbb{Z}. \tag{9}$$

Equivalently, considering expectation values for a state:

$$\Phi_{\text{total}} \equiv \langle\Psi_{\text{total}}|\hat{\Phi}_{\text{Euler}}^{(\text{total})}|\Psi_{\text{total}}\rangle = 2\pi N.$$

3 Stabilization mechanisms (operator-level)

Direct satisfaction of (9) for a realistic system is nontrivial. We introduce a stack of operators/filters that act on the phase operators or on the numeric phase values to obtain stabilization.

3.1 Zeta modulation operator

Define $\hat{\zeta} \equiv \zeta(\frac{1}{2} + i\hat{T})$ acting in the metatime operator representation $\hat{T}$ (see Neutrinotime in provided material). Operationally, implement as a multiplicative filter:

$$\tilde{\phi} \;=\; \zeta(\sigma + i\tau)\,\phi, \tag{10}$$

where (σ,τ) are chosen per physical scaling; numerical experiments in this manuscript use $\sigma = 0.6$ as an example.

3.2 Discrete Collatz-like iteration (operator $\hat{C}$)

We define a discrete map $\hat{C}$ acting on scalar phase values ϕ (working mod 2π):

$$C(\phi) = \begin{cases} \phi/2, & |\phi| < \pi, \\ 3\phi + \varepsilon, & |\phi| \geq \pi, \end{cases}$$

implemented together with reduction modulo 2π. The map is used iteratively to probe discrete attractors on the $[0, 2\pi)$ circle.

3.3 TwinPrimes check and grid

A coarse discrete grid of "anchor" indices is defined by mapping scaled phases to integer grid via α:

$$n_\alpha(\phi) = \mathrm{round}(\alpha\phi).$$

When $n_\alpha(\phi)$ matches an element of a chosen anchor set (e.g. indices that correspond to twin prime positions in a chosen enumeration), the iteration may be halted or the state marked as "anchored".

3.4 Ramanujan sums as final discretizer

Ramanujan sums provide a number-theoretic discretization operator for phases:

$$c_q(n) = \sum_{\substack{a=1 \\ (a,q)=1}}^{q} e^{2\pi i a n/q}. \tag{11}$$

We use these sums operationally to map continuous phase vectors to a q-grid in a way compatible with periodicity and arithmetical structure. In practice, one chooses small q (e.g. $q = 5$) to realize a coarse but robust discretization.

3.5 Fractal stability filter and Operator Intention

A fractal filter F_{fractal} identifies parameter-space regions with stable iterations (Mandelbrot-like set). The Operator of Intention $\hat{I}$ is a minimal, controlled displacement operator on the Hamiltonian parameters (energies E_n or path in parameter space) used only when the system lies close to but outside a stable basin. Mathematically:

$$\phi \mapsto \phi + \hat{I}[\phi], \qquad \|\hat{I}\| \ll \|\phi\|.$$

We emphasize keeping $\hat{I}$ minimal and physically interpretable (e.g. small change in adiabatic path, tiny potential perturbation).

3.6 White Threads / Holonomic transfer operator

Define $\hat{W}$ as an operator that models instantaneous, holonomic (nonlocal) synchronization of phase variables across microstates via entanglement-mediated transfer:

$$\hat{W} : \{\phi^{(n)}\} \mapsto \{\phi^{(n)} + \Delta^{(n)}\},$$

subject to global constraint and conservation laws. In the analogy, $\hat{W}$ implements the White Thread mechanism (instantaneous phase information redistribution), and in the BH/WH picture it models Hawking-like release of entangled-phase information.

3.7 Full stabilized operator

Concise operator composition:

$$\hat{\Phi}_{\text{Euler}}^{(\text{stable})} = \text{Ramanujan} \circ \text{TwinPrimesCheck} \circ \hat{C} \circ \hat{\zeta}(\hat{\Phi}_{\text{Euler}}^{(\text{total})}), \tag{12}$$

optionally augmented by F_{fractal} and $\hat{I}$, and monitored by $\hat{W}$.

4 Emergence vs arbitrariness

We adopt the following operational distinction:

- **Emergent:** quantities/structures that follow from dynamics, topology or number-theoretic selectors (Euler-Berry, Zeta, Ramanujan grids, White Threads as emergent entanglement patterns).
- **Arbitrary (numerical / set-up):** initial choices like specific initial energies E_n, choice of α for TwinPrimes grid, small $\hat{I}$ corrections introduced for numerics.

The manuscript keeps arbitrary choices minimal and documents them explicitly in numerical sections.

5 Numerical worked example: 3-level system (full pipeline)

This section reproduces the stepwise numeric pipeline used during model development; the values are those used in the discussions.

5.1 Initial data (units: $\hbar = 1$, $T = 1$)

Level	E_n	γ_B	ϕ_D	γ_P	(Zak, AC set to 0)
1	1.05	$\pi/3 \approx 1.047$	$\pi/6 \approx 0.524$	$\pi/12 \approx 0.262$	
2	1.48	$\pi/2 \approx 1.571$	$\pi/4 \approx 0.785$	$\pi/6 \approx 0.524$	
3	0.82	$\pi/6 \approx 0.524$	$\pi/3 \approx 1.047$	$\pi/4 \approx 0.785$	

Table 1: Initial parameters for the 3-level example.

5.2 Step 1: compute raw total phases (per level)

Define

$$\phi_{\text{total}}^{(i)} = \phi_{\text{dyn}}^{(i)} + \gamma_B^{(i)} + \phi_D^{(i)} + \gamma_P^{(i)}.$$

With $\phi_{\text{dyn}}^{(i)} = -E_i$ (single-unit time):

$$\begin{aligned}
\phi_{\text{total}}^{(1)} &\approx 1.047 - 1.05 + 0.524 + 0.262 = 0.783 \text{ rad}, \\
\phi_{\text{total}}^{(2)} &\approx 1.571 - 1.48 + 0.785 + 0.524 = 1.400 \text{ rad}, \\
\phi_{\text{total}}^{(3)} &\approx 0.524 - 0.82 + 1.047 + 0.785 = 1.536 \text{ rad}.
\end{aligned}$$

Global sum $\Phi_0 \approx 3.719$ rad (mod $2\pi \approx 3.719$), not equal to $2\pi N$.

5.3 Step 2: Zeta modulation

Apply $\zeta(\sigma)$ with $\sigma = 0.6$; numerically $\zeta(0.6) \approx -1.07235$ (value chosen in examples).

$$\tilde{\phi}^{(i)} = \zeta(0.6)\, \phi^{(i)}_{\text{total}}.$$

Numerically:

$$\tilde{\phi}^{(1)} \approx -0.840, \quad \tilde{\phi}^{(2)} \approx -1.500, \quad \tilde{\phi}^{(3)} \approx -1.646.$$

Global sum ≈ -3.986 rad $\equiv 2.297$ rad (mod 2π).

5.4 Step 3: Collatz-like iteration (mod 2π)

Apply $C(\cdot)$ with $\phi/2$ for $|\phi| < \pi$. First two iterations shown in the conversation; after two iterations we observed no immediate convergent hit to $2\pi N$. Iteration values (rounded as in discussion) were:

$$\phi^{(i)}_{k+1} = C(\phi^{(i)}_k), \qquad (\text{mod } 2\pi).$$

In numerical practice, always keep mod reduction after each step.

5.5 Step 4: TwinPrimes grid and Ramanujan final discretization

Choose scale $\alpha = 1.5$ and compute $n_\alpha = \text{round}(\alpha\phi)$. Compare to anchor set (e.g. indices corresponding to small twin primes). When no immediate anchor match exists, apply small fractal-guided $\hat{I}$ adjustments.

Then apply Ramanujan-filter with small q (e.g. $q = 5$) to finalize discretization. Effectively map continuous phase to nearest allowed discrete value modulo 2π.

5.6 Step 5: minimal fractal correction + Operator Intention

Compute required global correction

$$\Delta\Phi = 2\pi N - \sum_i \phi_i^{\text{current}},$$

and distribute $\Delta\Phi$ proportionally to $|\phi_i|$ as a minimal adjustment,

$$\Delta\phi_i = \Delta\Phi \cdot \frac{|\phi_i|}{\sum_j |\phi_j|}.$$

Apply only when system lies near a fractal stability basin; keep adjustments $\ll$ phases.

5.7 Outcome for the example (summary)

Following the pipeline (Zeta $\to$ Collatz $\to$ TwinPrimes $\to$ Ramanujan $\to$ minimal fractal $\hat{I}$), the global phase can be brought to $2\pi N$ with very small $\hat{I}$ corrections. The approach converts continuous phase data into a stable discrete configuration while preserving topological winding numbers (verified by White Threads tests).

6 Numerical worked example: stylized Sagittarius A* microstate grid

We present a stylized microstate model for Sagittarius A* using five representative microstates. The purpose is to illustrate scalability of the pipeline rather than to claim astrophysical precision.

6.1 Parameters

Assume in model units each microstate has effective energy $E \sim 4.3 \times 10^6$ (scaled), and nominal phases as in the preparatory notes:

$$\gamma_B \sim \{1.5, 1.6, 1.4, 1.55, 1.48\},\ \phi_D \sim \{0.6, 0.7, 0.5, 0.65, 0.6\},\ \gamma_P \sim \{0.3, 0.4, 0.35, 0.32, 0.33\},\ \ldots$$

Applying the same sequence (Zeta modulation $\sigma = 0.6$, Collatz reduction, grid mapping $\alpha = 1.5$ and Ramanujan discretization with $q = 5$) yields a stable discrete grid of anchor indices for all microstates. White Threads then interconnect anchors, modeling instantaneous entanglement-mediated transfer (the analogue of Hawking emission channels).

6.2 Interpretation

In this construction:

- the BH is a large, topologically bound ensemble of phases (microstate bundle);
- White Threads correspond to phase-entanglement channels that can transfer coherent phase information to the external domain in a manner compatible with the Euler constraint;
- the pipeline is scale-invariant in design (operators act analogously at micro and macro scales).

7 Ontological and physical implications

7.1 BH/WH duality

The operator-level duality is:

$$\hat{B} \equiv \sum_i |\psi_i\rangle\langle\psi_i| \quad \text{(binding information inside horizon)},$$

$$\hat{W} \equiv \text{(White Threads operator)} \quad \text{(holonomic information release)}.$$

Both operators are subject to global Euler constraint (they operate on the same phase algebra), and $\hat{W}$ provides the mechanism by which BH microstate phase information can be made accessible in the exterior, consistent with a topological phase accounting.

7.2 White Threads ≈ Hawking radiation (analogy)

Interpreting White Threads as phase-entanglement channels yields an alternative viewpoint on Hawking radiation: radiation is an emergent, coherent release of entangled-phase information constrained by global phase topology (Euler constraint), filtered by Zeta structure and discretized by number-theoretic anchors.

8 Emergent vs Arbitrary: recapitulation

The pipeline separates:

- genuinely emergent structures: Euler-Berry, Zeta filtering, fractal basins, Ramanujan discretization and White Threads (as entanglement-mediated synchronizers);
- practical/numerical choices: initial E_n, tuning parameter α, small $\hat{I}$ corrections, choice of q for Ramanujan sums.

9 Pseudocode / algorithmic pipeline

```
Given microstates {n} with energies E_n and phase components:
 for each n:
   compute phi_total[n] = phi_dyn + gamma_B + phi_D + gamma_P + phi_Zak + phi_AC
 apply zeta: phi_z[n] = zeta(sigma)*phi_total[n]
 iterate collatz: phi_c[n] = C(phi_z[n]) (mod 2pi) until stopping criteria
 map to integer grid: idx[n] = round(alpha*phi_c[n])
 check anchor: if idx[n] in TwinPrimesGrid then anchor[n]=true
 apply Ramanujan sum discretizer: phi_r[n] = RamanujanFilter(phi_c[n])
 if global sum != 2pi*N within tol:
   compute DeltaPhi = 2pi*N - sum(phi_r)
   distribute minimal fractal-guided corrections DeltaPhi_i
 verify topology (white threads / winding numbers)
 output stable configuration {phi_final[n]}
```

10 Conclusions and next steps

We provided a fully specified, modular formalism that unifies known phase contributions into a single global phase constraint and proposed a set of stabilization operators to achieve topological and numeric stability. Numerical examples demonstrate feasibility for small ensembles and stylized BH microstate grids. Next steps for rigorous development:

- implement the pipeline in reproducible code (Python/Julia/Matlab) with parameter sweeps over $\zeta(\sigma)$, α, q and fractal maps;
- analyze sensitivity and robustness (measure of minimal $\hat{I}$ needed, fractal basin sizes);
- design analogue experiments (SSH photonic lattices, BEC solitons) to test Euler-Berry constraints and White Thread signatures.

A Appendix A: numeric tables (3-level example stepwise)

Step	Level 1 (rad)	Level 2 (rad)	Level 3 (rad)	Global sum (rad)
Raw total	0.783	1.400	1.536	3.719
Zeta×	-0.840	-1.500	-1.646	-3.986
Collatz(1)	-0.4205	-0.750	-0.823	-1.994
Collatz(2)	-0.210	-0.375	-0.412	-0.997
Fractal+I	+0.403	+0.450	+0.600	+1.453
Ramanujan final (example)	0.628	0.858	0.504	1.990
Minimal correction -> 2pi	(distributed)			6.283

Table 2: Representative numbers from pipeline (rounded).

B Appendix B: notation and conventions

All phases in radians. $\hbar = 1$, $T = 1$ unless explicitly stated. "mod 2π" reduction applied after discrete maps. Zeta values taken as numerical examples for demonstration; if intended for proofs one must use precise operator spectrum of $\hat{T}$ and, where appropriate, spectral regularization.

References

This work builds upon well-known physical phase definitions (Berry, Aharonov–Bohm, Pancharatnam, Zak, Aharonov–Casher), number-theory constructs (Ramanujan sums, twin primes) and standard fractal iteration theory (Mandelbrot). For context and formal details, the reader should consult the canonical sources for Berry phase, Aharonov–Bohm effect, and standard texts on Riemann zeta and Ramanujan sums. Specific project notes available
https://github.com/AdrianLipa90/Metatime